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Article

# Evaluating Parametric and Non-Parametric Quantile Delta Mapping for CMIP6 Precipitation Bias Correction: Independent Validation in Coastal Thailand

Surasit Punyawansiri<sup>1\*</sup>

1 Office of Water Management and Hydrology, Royal Irrigation Department, Dusit, Bangkok 10300, Thailand

\* Correspondence: Surasit.irri@gmail.com

**Abstract:** Bias correction of climate-model precipitation is often assessed in the calibration period, which can overstate transferability. This study compared annual and month-wise parametric and non-parametric quantile delta mapping (QDM) for daily rainfall from five CMIP6 models at 12 stations in Prachuap Khiri Khan, Thailand. Transfer functions were fitted in 1981-2000 and frozen before independent validation in 2001-2014. Parametric candidates were selected by maximum-likelihood Akaike information criterion, and uncertainty was evaluated with 2,000 paired year-block bootstrap replicates. Month-wise methods were consistently stronger than annual variants. With the pre-specified five-times calibration-maximum cap, parametric monthly QDM reduced mean monthly RMSE from 116.20 to 105.06 mm and increased mean monthly Kling-Gupta efficiency from 0.229 to 0.337; non-parametric monthly QDM yielded 109.45 mm and 0.329, respectively. The paired median difference between the two monthly methods was nearly zero for KGE (0.0002), and the preferred method changed in the uncapped sensitivity run. Non-parametric annual QDM gave the lowest mean absolute 99th-percentile bias (24.79% versus 52.98% raw). All variants improved wet-day distributional agreement, reducing mean Kolmogorov-Smirnov D to 0.098-0.141 from 0.201. The evidence supports month-wise calibration but not a universal parametric or empirical winner; parametric gains require transparent tail safeguards, whereas empirical monthly QDM is the more assumption-lean default.

**Keywords:** bias correction; quantile delta mapping; CMIP6; independent validation; Thailand

## 1. Introduction

Daily rainfall controls water-resources planning, flood and drought risk, agricultural operations, and climate adaptation in Southeast Asia. CMIP6 simulations provide physically based climate information, but precipitation biases remain because of coarse resolution and imperfect representations of convection, topography, and regional circulation [1,2]. These limitations are especially consequential in Prachuap Khiri Khan, where monsoon seasonality, a narrow coastal plain, and the Tenasserim Range generate strongly intermittent and skewed rainfall.

Quantile mapping is widely used to align simulated and observed distributions [3,4,6,7]. Quantile Delta Mapping (QDM) extends this family by preserving modelled relative changes at each quantile [5], while scaled distribution mapping provides a related trend-preserving approach [8]. Method performance nevertheless depends on reference uncertainty, spatial mismatch, dry-day treatment, and the statistical representation of extremes [9,10]. Recent global and multi-region studies likewise show that the preferred method varies with climate, terrain, variable, and impact metric [10,14,15].

Three unresolved issues motivate the present comparison. First, evaluation in the calibration period measures construction-dependent agreement and can overstate out-of-sample performance. Second, univariate quantile mapping adjusts marginal distributions but cannot reconstruct missing event timing or wet-spell persistence [11]. Third, finite samples make upper-tail behaviour sensitive to extrapolation and parameterization [12,13]. These issues are particularly relevant when monthly stratification improves seasonality but reduces the wet-day sample available for fitting.

This study therefore compares four multiplicative QDM variants: non-parametric annual, non-parametric monthly, parametric annual, and parametric monthly. All observation-dependent quantities are fitted in 1981-2000, frozen, and applied to 2001-2014. The primary endpoint is validation-period monthly Kling-Gupta efficiency (KGE); secondary endpoints characterize distributional fidelity, upper-tail bias, seasonal structure, and temporal correspondence. The objective is to determine which conclusions are transferable, which depend on numerical safeguards, and which uses of the corrected rainfall remain defensible.

## 2. Materials and Methods

### 2.1 Study area, observations, and CMIP6 simulations

Prachuap Khiri Khan is a narrow province between the Gulf of Thailand and the Tenasserim Range. The analysis used daily rainfall at 12 Thai Meteorological Department gauges from 1 January 1981 through 31 December 2014 (12,418 days). The station codes and coordinates in the Hydro-Informatics Institute/Royal Irrigation Department project archive match the official 12-station inventory reported for the province [27]. The retained period is a common 1981-2014 subset of that longer gauge archive.

All stations had complete daily records. Declared missing codes were masked before negative-value handling. Negative values between -0.1 and 0 mm day<sup>-1</sup> were set to zero, and values below -0.1 mm day<sup>-1</sup> were treated as missing; no such corrections were required in the observation file used here. Annual totals were screened with the Pettitt change-point test at  $\alpha = 0.05$  [22]. No station rejected the null hypothesis ( $p = 0.059$ -1.000), which is evidence of no detected change point rather than proof of homogeneity. No homogenisation was applied.

The CMIP6 input comprised daily precipitation (pr) from the historical experiment for ACCESS-ESM1-5, CESM2, CanESM5, EC-Earth3, and MIROC6. The source, experiment, member, grid, variable, frequency, and date range are encoded in the analysis-ready filenames and reported in Table 2; model descriptions follow the corresponding

publications [16-20]. These identifiers were cross-checked against the federated Earth System Grid Federation (ESGF) catalogue on 28 August 2026. Because ESGF datasets can be replicated across data nodes, the complete CMIP6 facets are used as the stable source description rather than a transient mirror endpoint. Values were already in mm day<sup>-1</sup>; the workflow checks units and would multiply flux units in kg m<sup>-2</sup> s<sup>-1</sup> by 86,400. The station-series files contain two or three distinct model series across the 12 gauges, consistent with nearest native-grid extraction and shared grid cells. A machine-readable manifest records the exact derivative filenames, byte sizes, and SHA-256 checksums used in the analysis.

Standard-calendar models and observations contain 7,305 calibration and 5,113 validation days. CESM2 and CanESM5 use a 365-day calendar, with 7,300 and 5,110 days. Analysis used dates common to each model and the observations; 29 February was omitted for 365-day models and was not interpolated. The 60 model-station combinations are therefore paired evaluation units, not 60 independent spatial experiments, because stations can share model grid cells and the models share components and development lineages.

**Table 1.** Rain-gauge station characteristics for 1981-2014.

| Station | Lat. (°N) | Long. (°E) | Elev. (m) | Missing (%) | Annual rainfall (mm) | SDII (mm day <sup>-1</sup> ) | Max. daily rainfall (mm day <sup>-1</sup> ) | Lag-1 ACF |
|---------|-----------|------------|-----------|-------------|----------------------|------------------------------|---------------------------------------------|-----------|
| 500001  | 12.37     | 99.95      | 9.1       | 0.0         | 984                  | 12.0                         | 170.0                                       | 0.21      |
| 500002  | 11.18     | 99.55      | n/a       | 0.0         | 1,376                | 11.1                         | 155.0                                       | 0.22      |
| 500003  | 12.02     | 99.90      | n/a       | 0.0         | 1,100                | 13.1                         | 181.3                                       | 0.19      |
| 500004  | 11.47     | 99.65      | n/a       | 0.0         | 1,106                | 11.4                         | 152.0                                       | 0.21      |
| 500005  | 11.60     | 99.68      | 5.0       | 0.0         | 1,147                | 6.4                          | 143.5                                       | 0.41      |
| 500006  | 11.62     | 99.67      | 23.0      | 0.0         | 1,203                | 6.5                          | 201.1                                       | 0.35      |
| 500007  | 11.90     | 99.78      | 21.3      | 0.0         | 1,134                | 6.2                          | 176.6                                       | 0.35      |
| 500008  | 12.53     | 99.55      | 130.3     | 0.0         | 1,206                | 5.1                          | 68.0                                        | 0.66      |
| 500009  | 12.22     | 99.87      | 14.4      | 0.0         | 1,177                | 5.2                          | 54.2                                        | 0.63      |
| 500201  | 11.83     | 99.83      | n/a       | 0.0         | 1,108                | 12.1                         | 296.9                                       | 0.25      |
| 500202  | 12.59     | 99.96      | n/a       | 0.0         | 944                  | 11.8                         | 298.5                                       | 0.26      |
| 500301  | 12.58     | 99.73      | 101.8     | 0.0         | 1,065                | 11.4                         | 226.0                                       | 0.25      |

Note: n/a indicates that elevation metadata were absent from the supplied station file. SDII is mean rainfall on wet days ( $\geq 1.0$  mm day<sup>-1</sup>), and ACF is lag-1 autocorrelation.

**Table 2.** CMIP6 historical simulations and metadata recovered from the analysis-ready files.

| Model         | Institution         | Variant  | Grid | Resolution (degrees) | Calendar | Days   | Distinct series | Reference |
|---------------|---------------------|----------|------|----------------------|----------|--------|-----------------|-----------|
| ACCESS-ESM1-5 | CSIRO               | r1i1p1f1 | gn   | 1.875 × 1.25         | standard | 12,418 | 2               | [20]      |
| CESM2         | NCAR                | r1i1p1f1 | gn   | 1.25 × 0.90          | 365-day  | 12,410 | 3               | [17]      |
| CanESM5       | CCCma               | r1i1p1f1 | gn   | 2.8 × 2.8            | 365-day  | 12,410 | 2               | [16]      |
| EC-Earth3     | EC-Earth Consortium | r1i1p1f1 | gr   | 1.125 × 1.125        | standard | 12,418 | 3               | [18]      |

| Model  | Institution      | Variant  | Grid | Resolution (degrees) | Calendar | Days   | Distinct series | Reference |
|--------|------------------|----------|------|----------------------|----------|--------|-----------------|-----------|
| MIROC6 | MIROC Consortium | r1i1p1f1 | gn   | 1.4 x 1.4            | standard | 12,418 | 2               | [19]      |

Note: Distinct series is the number of unique daily station-series signatures across 12 gauges and indicates shared native grid cells; it is not a formal effective sample size.

## 2.2 Split-sample protocol and wet-day adaptation

Calibration covered 1981-2000 and independent validation covered 2001-2014. Every observed threshold, empirical distribution, and parametric calibration distribution was estimated from calibration data only and then frozen. Validation observations were used solely to calculate evaluation metrics.

Wet-day occurrence and positive intensity were corrected separately. A day was wet when observed rainfall was at least 1.0 mm day<sup>-1</sup>. For each model, station, and calibration group,  $p_{\text{dry}}$  was the fraction of observed calibration days below 1.0 mm day<sup>-1</sup>. The model threshold was the empirical  $p_{\text{dry}}$  quantile of all model calibration values. Values at or below that threshold were set to zero; with ties, this rule ensures that the adapted model wet-day count does not exceed the observed count. The threshold was reused unchanged in validation. Annual variants used one threshold per model-station pair, whereas monthly variants used calendar-month-specific thresholds.

A minimum of 30 wet days was required for a monthly transfer function. Of 720 model-station-month blocks, 100 dry-season blocks did not meet this threshold. These blocks were not discarded or set to zero; they used the corresponding annual transfer function fitted from calibration data. This fallback was declared uniformly and recorded in the diagnostics.

## 2.3 Multiplicative QDM

For a model value  $x$  at time  $t$  in target period  $p$ , multiplicative QDM followed Cannon et al. [5]:

$$\tau(t) = F_{m,p}[x_{m,p}(t)] \quad (1)$$

$$\Delta(t) = x_{m,p}(t) / F_{m,h}^{-1}[\tau(t)] \quad (2)$$

$$\hat{x}(t) = F_{o,h}^{-1}[\tau(t)] \times \Delta(t) \quad (3)$$

Here  $F_{m,p}$  is the model cumulative distribution in the target period;  $F_{m,h}^{-1}$  and  $F_{o,h}^{-1}$  are model and observed calibration-period quantile functions;  $\tau$  is non-exceedance probability;  $\Delta$  is the modelled relative-change ratio; and  $\hat{x}$  is corrected rainfall. The target-period probability uses model values only and therefore does not violate the observation holdout. Retaining  $\Delta$  distinguishes QDM from ordinary quantile mapping.

$\Delta$  was bounded a priori to [0.2, 5.0] to prevent division by very small historical quantiles. For the primary parametric analysis, corrected rainfall was capped at five times the maximum observed wet-day rainfall in the corresponding calibration block. The cap was applied after transformation. Sensitivity runs used a factor of 1.5 and no cap; all clipping and capping events were counted.

## 2.4 Parametric and non-parametric variants

Non-parametric variants used empirical cumulative distribution functions with plotting positions  $(i - 0.5)/n$  and linear interpolation. Annual variants pooled all months, whereas monthly variants fitted each calendar month separately.

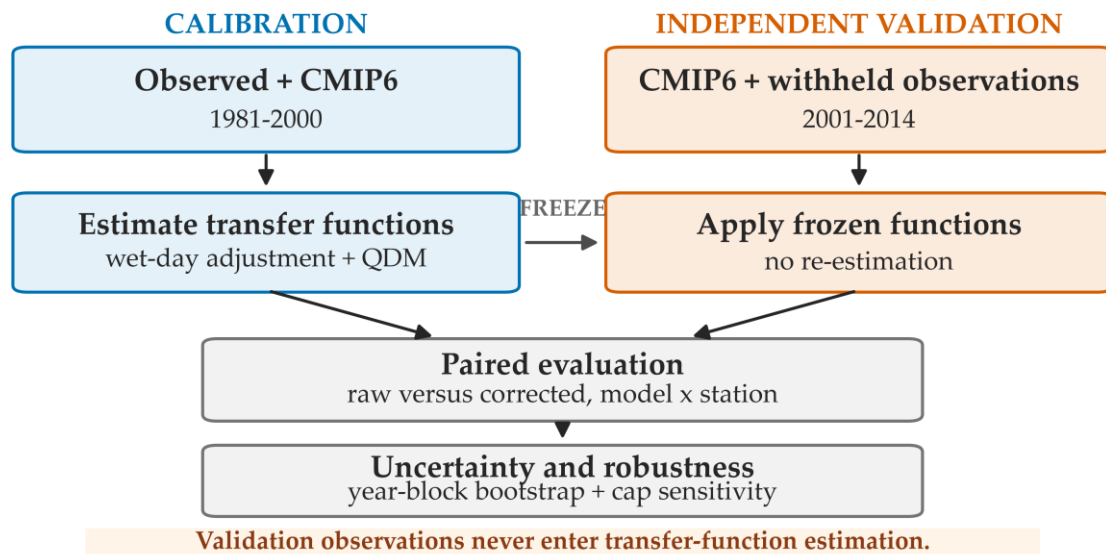
Parametric variants fitted gamma, generalized extreme value, Weibull minimum, lognormal, and Pearson type III candidates to wet-day intensities by maximum likelihood. Gamma, Weibull, and lognormal location parameters were fixed at zero; the generalized extreme value and Pearson type III location parameters were estimated.  $AIC = 2k - 2 \ln(L)$ , where  $k$  is the number of freely estimated parameters and  $L$  is maximized likelihood. The minimum-AIC candidate was selected deterministically; non-finite parameters or likelihoods disqualified a candidate. The Kolmogorov-Smirnov statistic was stored as a fit diagnostic. The generalized extreme value distribution was used as a flexible whole-distribution candidate, not as a claim that daily wet-day rainfall is a block-maximum sample. Calculations used Python 3.12.13 and SciPy 1.18.1 [24].

## 2.5 Evaluation endpoints and uncertainty

The primary endpoint was monthly KGE [21], selected because it combines correlation, variability, and mean bias. Secondary endpoints were the Kolmogorov-Smirnov statistic for wet-day distributions, absolute relative bias of the 99th wet-day percentile, monthly RMSE, monthly correlation, and monthly-climatology correlation. KGE and correlations are better when larger; the remaining endpoints are better when smaller. The phrase strongest overall is used only for the highest mean primary endpoint, while discordant secondary endpoints are reported separately.

Uncertainty in corrected-minus-raw differences was estimated separately for each model-station combination with 2,000 paired year-block bootstrap replicates and seed 42. Validation monthly totals were arranged as a 14-year by 12-month matrix. Fourteen years were sampled with replacement in each replicate, and the same year indices were applied to observations, raw simulations, and all corrected variants. The 95% interval was the 2.5th and 97.5th percentiles. Two-sided  $p$  values used a plus-one correction, and Benjamini-Hochberg adjustment controlled the false-discovery rate at 0.05 within each method-metric family [23]. Because combinations are spatially and structurally dependent, network summaries remain descriptive and are not treated as 60 independent experiments.

**Figure 1.** Fit-freeze-apply framework. Observation-dependent transfer functions were estimated in 1981-2000 and applied without re-estimation in 2001-2014; uncertainty used paired year-block bootstrap resampling.



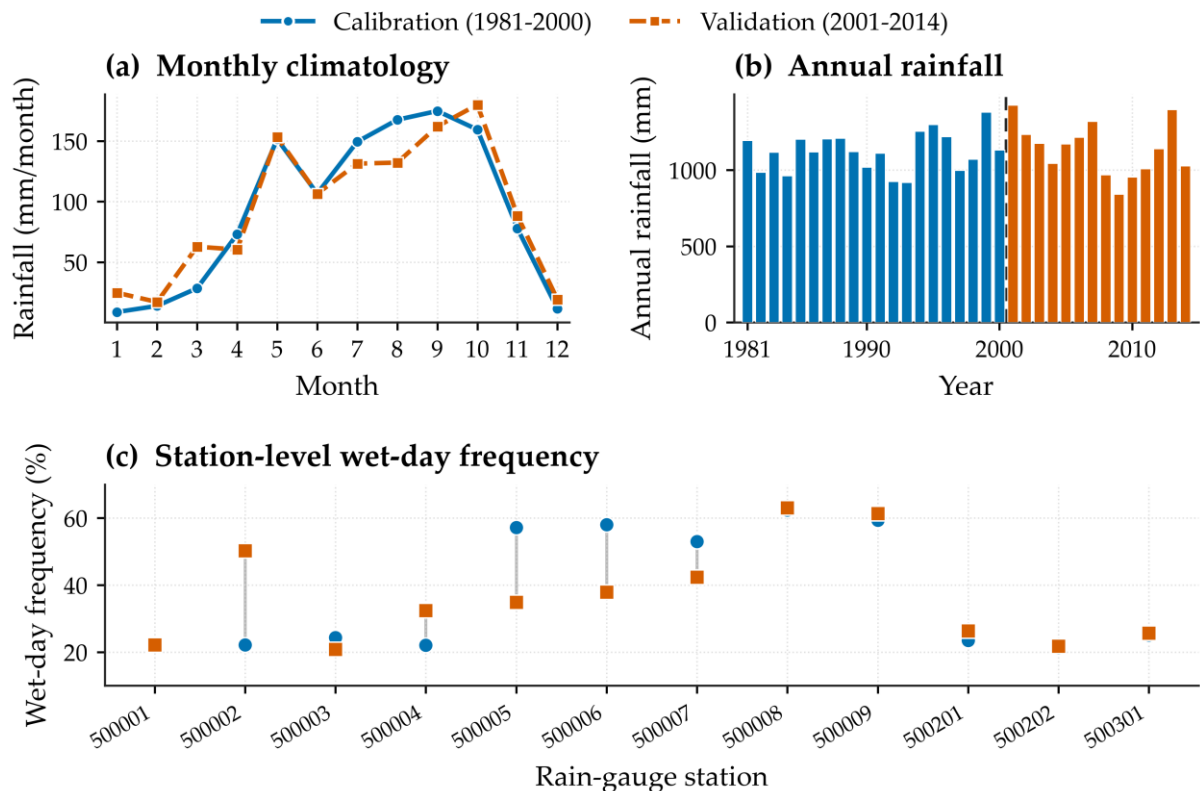
### 3. Results and Discussion

#### 3.1 Observed rainfall and model context

Mean annual observed rainfall ranged from 944 to 1,376 mm among stations, and maximum daily rainfall ranged from 54.2 to 298.5 mm day<sup>-1</sup>. Calibration and validation retained the same broad bimodal seasonal cycle, but several stations changed wet-day frequency between periods (Figure 2). This non-stationarity makes the holdout test more demanding than same-period evaluation.

The five model files represented only two or three distinct native-grid series across the 12 stations (Table 2). Consequently, averaging across all 60 combinations would overstate spatial replication. Results are therefore reported as paired model-station distributions and counts, with this dependence carried into interpretation.

**Figure 2.** Observed rainfall in calibration and validation: (a) monthly climatology, (b) annual rainfall, and (c) paired station-level wet-day frequencies. Connecting lines in panel (c) show the change at each gauge.



### 3.2 Independent validation performance

Month-wise calibration produced the clearest gains (Table 3; Figure 3). P monthly had the highest mean primary endpoint, increasing monthly KGE from 0.229 to 0.337, and the lowest monthly RMSE, decreasing it from 116.20 to 105.06 mm. NP monthly was close, with KGE 0.329 and RMSE 109.45 mm. Both monthly methods increased monthly-climatology correlation to at least 0.864 and improved that metric in 59 of 60 combinations.

The paired bootstrap did not support a broad parametric advantage over the empirical monthly method. For P monthly minus NP monthly, the median combination-level KGE difference was 0.0002, with 30 combinations favouring each method. The median RMSE difference was -1.45 mm, with P monthly lower in 36 combinations and NP monthly lower in 24. Thus, the aggregate ranking is driven by heterogeneous station-model responses rather than uniform dominance.

Upper-tail performance followed a different ordering. NP annual reduced mean absolute q99 bias from 52.98% to 24.79%, compared with 27.38% for NP monthly and 32.54% for P monthly. Pooling months provides a larger empirical wet-day sample and therefore more stable upper quantiles, whereas month-wise fitting directly resolves seasonality at the cost of smaller samples. No method was best on every endpoint.

Annual variants were weaker for monthly performance: NP annual increased mean RMSE to 124.74 mm and reduced mean KGE to 0.203, while P annual yielded 123.72 mm

and 0.152. Distributional correction alone therefore does not guarantee better month-to-month agreement. This is consistent with the limitation that univariate mapping cannot reconstruct event sequencing absent from free-running climate simulations [11].

**Table 3.** Independent validation means across 60 model-station combinations.

| Metric                     | Raw    | NP annual | NP monthly | P annual | P monthly | Best mean  |
|----------------------------|--------|-----------|------------|----------|-----------|------------|
| Mean absolute PBIAS (%)    | 19.99  | 13.47     | 13.81      | 22.13    | 15.48     | NP annual  |
| Kolmogorov-Smirnov D       | 0.201  | 0.106     | 0.101      | 0.141    | 0.098     | P monthly  |
| Mean absolute q99 bias (%) | 52.98  | 24.79     | 27.37      | 44.86    | 32.54     | NP annual  |
| Monthly RMSE (mm)          | 116.20 | 124.74    | 109.45     | 123.72   | 105.06    | P monthly  |
| Monthly MAE (mm)           | 78.15  | 81.06     | 70.47      | 80.46    | 67.40     | P monthly  |
| Monthly KGE                | 0.229  | 0.203     | 0.329      | 0.152    | 0.337     | P monthly  |
| Monthly correlation r      | 0.342  | 0.328     | 0.425      | 0.342    | 0.428     | P monthly  |
| Monthly climatology r      | 0.643  | 0.629     | 0.880      | 0.644    | 0.864     | NP monthly |

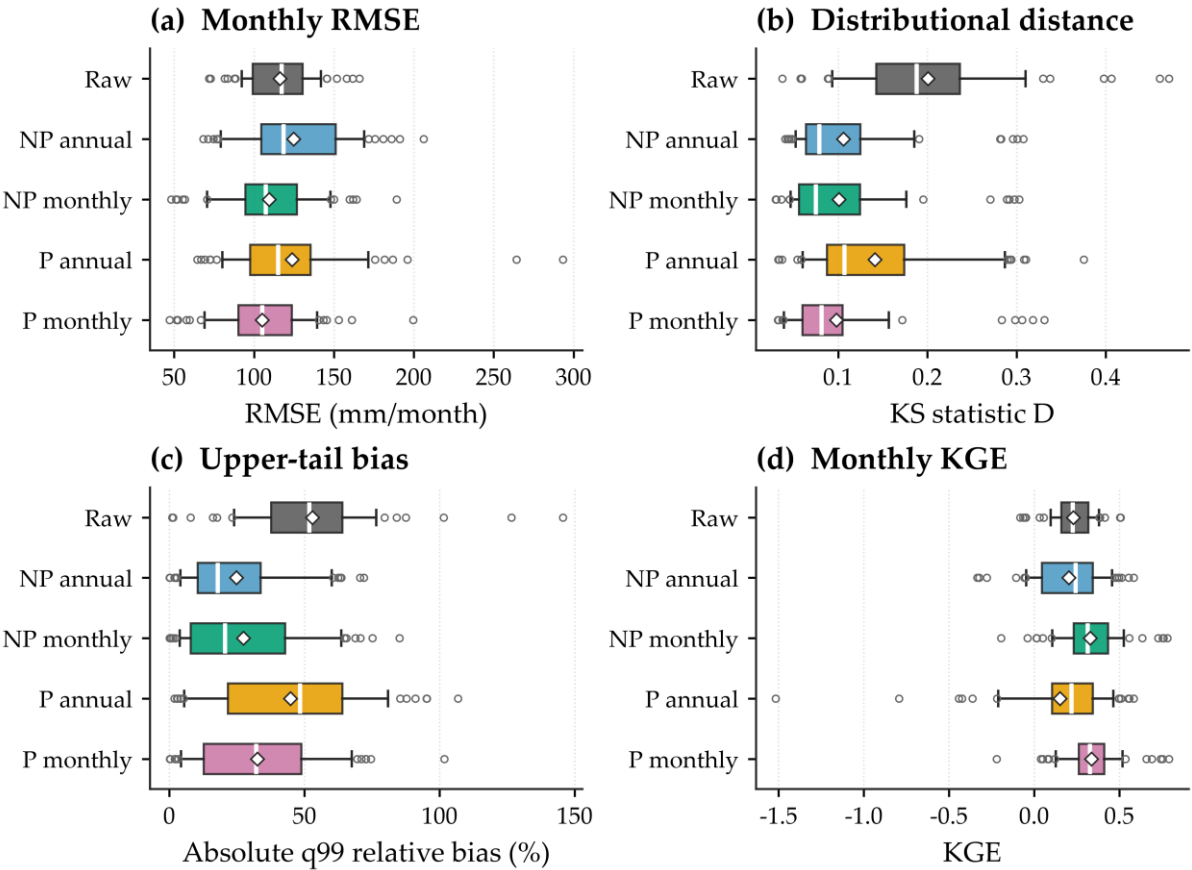
Note: PBIAS and q99 bias are summarized as absolute values. Best mean is descriptive; the 60 combinations are not independent.

**Table 4.** Paired year-block bootstrap summary for corrected-minus-raw monthly metrics.

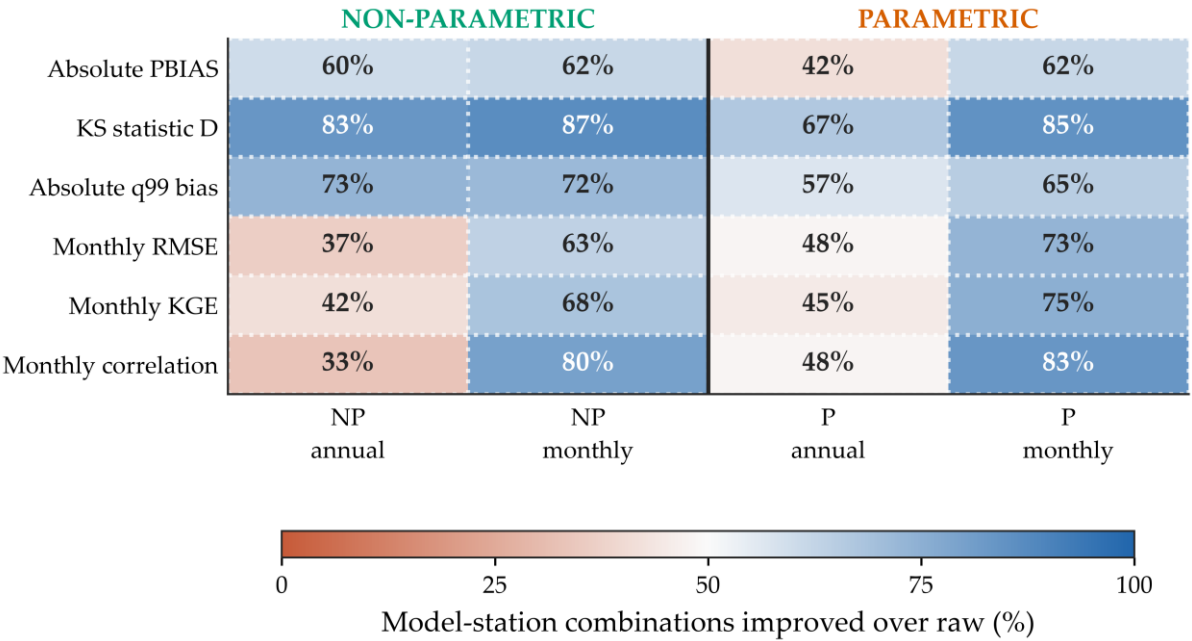
| Method     | Metric | Raw mean | Corrected mean | Median delta | Median 95% interval | Point improved | CI improved/worsened /overlap |
|------------|--------|----------|----------------|--------------|---------------------|----------------|-------------------------------|
| NP annual  | mRMSE  | 116.20   | 124.74         | +8.75        | [+3.33, +17.16]     | 22/60          | 19/32/9                       |
| NP annual  | mKGE   | 0.229    | 0.203          | -0.048       | [-0.126, +0.057]    | 25/60          | 14/20/26                      |
| NP monthly | mRMSE  | 116.20   | 109.45         | -5.57        | [-24.68, +10.13]    | 38/60          | 24/5/31                       |
| NP monthly | mKGE   | 0.229    | 0.329          | +0.071       | [-0.078, +0.264]    | 41/60          | 20/0/40                       |
| P annual   | mRMSE  | 116.20   | 123.72         | +0.73        | [-3.00, +4.67]      | 29/60          | 23/27/10                      |
| P annual   | mKGE   | 0.229    | 0.152          | -0.006       | [-0.118, +0.056]    | 27/60          | 16/19/25                      |
| P monthly  | mRMSE  | 116.20   | 105.06         | -11.07       | [-26.99, +2.54]     | 44/60          | 27/1/32                       |
| P monthly  | mKGE   | 0.229    | 0.337          | +0.095       | [-0.041, +0.261]    | 45/60          | 21/0/39                       |

Note: Interval bounds are the medians of 60 combination-specific bootstrap limits, not a confidence interval for the network median. For RMSE, negative delta is improvement; for KGE, positive delta is improvement.

**Figure 3.** Validation distributions for 60 model-station combinations per variant. Boxes show the interquartile range, whiskers show the 10th-90th percentiles, diamonds show means, and centre lines show medians; these are not bootstrap distributions.



**Figure 4.** Percentage of the 60 model-station combinations improved over raw CMIP6 simulations. Values above 50% indicate improvement for most paired combinations; percentages are not independent replicates.



### 3.3 Parametric selection, fallback, and tail sensitivity

Maximum-likelihood fitting changed the distribution-selection pattern relative to the earlier method-of-moments implementation. Pearson type III was selected for 55 of 60 observed annual fits and 610 of 620 eligible observed monthly fits. Model-side fits were more diverse: Pearson type III accounted for 496 monthly fits, generalized extreme value for 106, lognormal for 12, gamma for 3, and Weibull minimum for 3 (Table 5). All 620 eligible monthly blocks fitted successfully. The apparent deficits of 5 observed and 33 model fits in the earlier table were omitted gamma and Weibull categories, not optimization failures.

One hundred monthly blocks were ineligible because at least one calibration wet-day sample contained fewer than 30 values. These blocks clustered in dry-season months and used the annual fallback. This distinction is essential: the structural total remains 720, selection counts sum to the 620 eligible monthly fits, and fallback counts account for the remaining 100.

Tail safeguards affected the parametric ranking (Table 6). With the primary factor of 5.0, P monthly KGE was 0.337 versus 0.329 for NP monthly; without a cap, P monthly KGE decreased to 0.327. A stricter 1.5 factor capped 171 P-monthly validation values and improved mean RMSE to 101.91 mm, illustrating that aggressive truncation can improve central error metrics while imposing a strong assumption on extremes. The cap sensitivity therefore precludes a universal claim that parametric monthly QDM is superior.

The q99 metric changed little across the three parametric cap settings because most capped values were above the station-level 99th percentile. This does not make the cap inconsequential: a small number of altered maxima can materially affect return-level, IDF, or flood applications. Recent work similarly emphasizes that tail handling can behave differently inside and outside calibration and must be evaluated explicitly [12,13].

**Table 5.** Parametric fit accounting and numerical safeguards in validation.

| Method    | Eligible/structural | Observed selection                    | Model selection                                                  | Annual fallback | Fit failures | Capped | Clipped ratios |
|-----------|---------------------|---------------------------------------|------------------------------------------------------------------|-----------------|--------------|--------|----------------|
| P annual  | 60/60               | pearson3 55; lognorm 5                | pearson3 45; genextreme 15                                       | 0               | 0            | 3      | 247            |
| P monthly | 620/720             | pearson3 610; genextreme 5; lognorm 5 | pearson3 496; genextreme 106; lognorm 12; gamma 3; weibull_min 3 | 100             | 0            | 8      | 229            |

*Note: Monthly selection counts sum to 620 eligible blocks. The 100 ineligible blocks used annual calibration-only transfer functions. Clipped ratios combine lower- and upper-bound events.*

**Table 6.** Sensitivity of parametric validation metrics to the upper-tail cap factor.

| Cap factor    | P annual RMSE | P annual KGE | P monthly RMSE | P monthly KGE | P monthly  q99 bias  (%) | P annual capped | P monthly capped |
|---------------|---------------|--------------|----------------|---------------|--------------------------|-----------------|------------------|
| 1.5           | 122.01        | 0.167        | 101.91         | 0.355         | 32.60                    | 35              | 171              |
| 5.0 (primary) | 123.72        | 0.152        | 105.06         | 0.337         | 32.54                    | 3               | 8                |
| None          | 126.64        | 0.125        | 106.46         | 0.327         | 32.54                    | 0               | 0                |

*Note: The cap is factor  $\times$  the maximum observed wet-day rainfall in the relevant calibration block. None denotes the uncapped sensitivity run. NP results are invariant to this parameter.*

### 3.4 Implications and limitations

The model-specific raw results illuminate why a single correction function was insufficient. ACCESS-ESM1-5 and MIROC6 overestimated wet-day frequency by 0.227 and 0.232, respectively, and had positive mean PBIAS of 18.7% and 25.3%. EC-Earth3 instead had a negative mean PBIAS of -15.9% but the lowest raw monthly RMSE (92.30 mm) among the five models. Thus, similar aggregate errors arose from different combinations of event frequency, intensity, and seasonal phase. This heterogeneity agrees with Southeast Asian CMIP6 evaluations showing that rainfall skill reflects both monsoon circulation and model-specific parameterization, and that good large-scale circulation does not guarantee local rainfall skill [25,26].

A physically plausible interpretation follows from the province's narrow coastal geometry. Moist southwesterly flow encounters the Tenasserim Range along the western boundary, while Gulf moisture and the northeast monsoon contribute to late-year rainfall on the eastern coastal plain [27]. Coarse native grids cannot fully resolve this sharp mountain-coast gradient, local convection, or rain-shadow transitions. The especially large wet-frequency excess in ACCESS-ESM1-5 and MIROC6 is therefore consistent with overly frequent resolved or parameterized rainfall, whereas EC-Earth3's negative volume bias suggests a different compensation between frequency and intensity. These are process-informed interpretations, not causal attribution, because circulation, moisture-flux, and convection diagnostics were not analysed in this study.

The most stable result is the advantage of month-wise calibration for seasonal and monthly metrics. Annual pooling combines dry-season occurrence, southwest-monsoon rainfall, and the late-year coastal maximum in one transfer function. Month-wise QDM instead estimates separate wet-day thresholds and intensity distributions for each phase of the seasonal cycle, which better respects the changing atmospheric regime without claiming to simulate its dynamics. Both monthly methods consequently improved RMSE, KGE, monthly correlation, and climatology more consistently than annual calibration. The countervailing cost is a roughly twelve-fold reduction in sample size per transfer function, which explains the dry-season fallback and contributes to upper-tail uncertainty.

Method choice should follow application needs. NP monthly is an assumption-lean default for monthly water-balance and seasonal applications. NP annual is preferable when the 99th-percentile bias is the priority. P monthly can reduce monthly error further, but its advantage is conditional on the selected distributions and tail constraint. None of the variants should be interpreted as restoring observed daily event timing, spatial independence, or wet-spell persistence.

Four limitations remain. The study covers one province, 12 gauges, five historical runs, and a 14-year validation window. Shared native grid cells reduce spatial replication, and the analysis-ready CMIP6 products are station-matched derivatives rather than original NetCDF files with per-file tracking identifiers. The reported facets and checksum manifest identify the exact derivatives used, while independent recreation requires repeating the spatial extraction from the corresponding ESGF datasets. Finally, no future SSP experiment was analysed, so preservation of projected quantile change was implemented by the QDM formulation but not empirically evaluated in this historical holdout study. Future work

should test moving-window seasonal fits, explicit extreme-value tail models, circulation diagnostics, spatial dependence, and multiple future scenarios.

#### 4. Conclusions

Independent temporal validation shows that month-wise QDM is more transferable than annual pooling for monthly rainfall performance in Prachuap Khiri Khan. P monthly had the highest mean KGE and lowest mean RMSE under the pre-specified five-times cap, but its KGE advantage over NP monthly was negligible in paired bootstrap comparisons and reversed when the cap was removed. The defensible conclusion is therefore not that one distribution family universally wins, but that seasonal stratification drives the principal gain while parametric ranking remains tail-sensitive.

NP annual provided the strongest q99 correction, whereas both monthly methods best represented seasonal structure. All variants improved wet-day distributional agreement but did not reliably improve temporal correspondence. For operational use, NP monthly is the conservative default, NP annual is justified for upper-quantile emphasis, and P monthly should be used only with documented fitting, fallback, clipping, cap sensitivity, and application-specific extreme-value checks. The fit-freeze-apply protocol provides a more stringent basis for transferability assessment than same-period evaluation.

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**Data Availability Statement:** The rainfall observations are Thai Meteorological Department gauge records distributed through the Hydro-Informatics Institute/Royal Irrigation Department project archive; the official station inventory is documented in [27]. CMIP6 source datasets are discoverable through ESGF using the source, experiment, member, table, variable, and grid facets reported in Table 2. The reproducibility package contains the analysis code, exact input-file SHA-256 manifest, corrected series, metrics, bootstrap results, diagnostics, and publication figures. Redistribution of gauge observations remains subject to the data custodian's terms.

**Conflicts of Interest:** The author declares no conflict of interest.

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